Lab 08

Configuring IOS Intrusion Prevention System (IPS)

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8.1 Objective

The Objectives of this lab are:

- Enable IOS IPS.
- Configure logging.
- Modify an IPS signature.
- Verify IPS.

8.2 Background

Your task is to enable IPS on R1 to scan traffic entering the 192.168.1.0 network.

The server labeled Syslog is used to log IPS messages. You must configure the router to identify the syslog server to receive logging messages. Displaying the correct time and date in syslog messages is vital when using syslog to monitor the network. Set the clock and configure the timestamp service for logging on the routers. Finally, enable IPS to produce an alert and drop ICMP echo reply packets inline.

The server and PCs have been preconfigured. The routers have also been preconfigured with the following:

- Enable password: **ciscoenpa55**
- Console password: **ciscoconpa55**
- SSH username and password: **SSHadmin / ciscosshpa55**
- OSPF 101

Topology

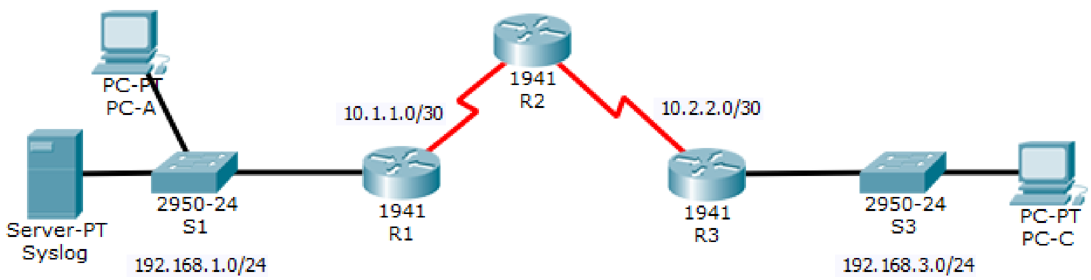


Figure 1: Topology

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/1
	S0/0/0	10.2.2.1	255.255.255.252	N/A	N/A
Syslog	NIC	192.168.1.50	255.255.255.0	192.168.1.1	S1 F0/2
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1	S1 F0/3
PC-C	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/2

Table 1: Addressing Table

Task 1: Configuring IOS Intrusion Prevention System

Step 1: Enable IOS IPS

Note: Within Packet Tracer, the routers already have the signature files imported and in place. They are the default xml files in flash. For this reason, it is not necessary to configure the public crypto key and complete a manual import of the signature files.

1. Enable the Security Technology package on **R1**.

```
R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#license boot module c1900 technology-package securityk9
```

2. Verify network connectivity
   a. Ping from **PC-C** to **PC-A**. The ping should be successful

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=15ms TTL=125
Reply from 192.168.1.2: bytes=32 time=2ms TTL=125
Reply from 192.168.1.2: bytes=32 time=2ms TTL=125
Reply from 192.168.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 15ms, Average = 5ms

C:\>
```

   b. Ping from **PC-A** to **PC-C**. The ping should be successful.

```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=2ms TTL=125
Reply from 192.168.3.2: bytes=32 time=2ms TTL=125
Reply from 192.168.3.2: bytes=32 time=11ms TTL=125
Reply from 192.168.3.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 11ms, Average = 4ms

C:\>
```

3. Create an IOS IPS configuration directory in flash.

   On **R1**, create a directory in flash using **mkdir** command. Name the directory **ipsdir.**

```
R1#mkdir ipsdir
Create directory filename [ipsdir]?
Created dir flash:ipsdir

R1#
```

4. Configure the IPS signature storage location.

   On **R1**, configure the IPS signature storage location to the directory you just created.

```
R1(config)#ip ips config location flash:ipsdir
```

5. Create an IPS rule.

   On **R1**, create an IPS rule name using **ip ips name** *name* command in global configuration mode. Name the IPS rule **iosips**.

```
R1(config)#ip ips config location flash:ipsdir
R1(config)#ip ips name iosips
R1(config)#
```

6. Enable logging. IOS IPS supports the use of syslog to send event notification. Syslog notification is enabled by default. If logging console is enabled, IPS syslog messages display.

   a. Enable syslog if it is not enabled.

```
R1(config)# ip ips notify log
```

   b. If necessary, use the **clock set** command from privileged EXEC mode to reset the clock.

```
R1(config)#ip ips notify log
R1(config)#logging 192.168.1.50
R1(config)#service timestamps log datetime msec
R1(config)#
```

c. Verify that the timestamp service for logging is enabled on the router using the **show run** command.

```
R1#show run | include timestamps
service timestamps log datetime msec
no service timestamps debug datetime msec
```

d. Send log messages to the syslog server at IP address 192.168.1.50

```
R1(config)#logging 192.168.1.50
```

7. Configure IOS IPS to use signature categories.

Retire the **all** signature category with the **retired true** command (all signatures within the signature release). Unretire the **IOS_IPS Basic** category with the **retired false** command.

```
R1(config)#ip ips signature-category
R1(config-ips-category)#category all
R1(config-ips-category-action)#retired true
R1(config-ips-category-action)#exit
R1(config-ips-category)#category ios_ips basic
R1(config-ips-category-action)#retired false
R1(config-ips-category-action)#exit
R1(config-ips-category)#exit
Do you want to accept these changes? [confirm]
Applying Category configuration to signatures ...
%IPS-6-ENGINE_BUILDING: atomic-ip - 288 signatures - 6 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 30 ms - packets for this engine will be scanned
```

8. Apply the IPS rule to an interface.

Apply the IPS rule to an interface with the **ip ips name** *direction* command in interface configuration mode. Apply the rule outbound on the G0/1 interface of **R1**. After you enable IPS, some log messages will be sent to the console line indicating that the IPS engines are being initialized.

Note: The direction **in** means that IPS inspects only traffic going into the interface. Similarly, **out** means that IPS inspects only traffic going out of the interface.

```
R1(config)#interface g0/1
R1(config-if)#ip ips iosips out
R1(config-if)#
*Mar 01, 00:11:49.1111:  %IPS-6-ENGINE_BUILDS_STARTED:  00:11:49 UTC Mar 01 1993
*Mar 01, 00:11:49.1111:  %IPS-6-ENGINE_BUILDING: atomic-ip - 3 signatures - 1 of 13 engines
*Mar 01, 00:11:49.1111:  %IPS-6-ENGINE_READY: atomic-ip - build time 8 ms - packets for this engine
will be scanned
*Mar 01, 00:11:49.1111:  %IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 8 mse
R1(config-if)#
```

Step 2: Modify the Signature

Note: For all configuration tasks, be sure to use the exact names as specified.

1. Change the event-action of a signature.

   Un-retire the echo request signature (signature 2004, subsig ID 0), enable it, and change the signature action to alert and drop.

```
R1(config)#ip ips signature-definition
R1(config-sigdef)#signature 2004 0
R1(config-sigdef-sig)#status
R1(config-sigdef-sig-status)#retired false
R1(config-sigdef-sig-status)#exit
R1(config-sigdef-sig)#engine
R1(config-sigdef-sig-engine)#event-action
R1(config-sigdef-sig-engine)#event-action p
R1(config-sigdef-sig-engine)#event-action produce-alert  d
R1(config-sigdef-sig-engine)#event-action produce-alert  drop-packet
```

2. Use show commands to verify IPS.

   Use the **show ip ips all** command to view the IPS configuration status summary. To which interfaces and in which direction is the **iosips** rule applied?

```
R1#show ip ips all
IPS Signature File Configuration Status
    Configured Config Locations: flash:ipsdir
    Last signature default load time:
    Last signature delta load time:
    Last event action (SEAP) load time: -none-

    General SEAP Config:
    Global Deny Timeout: 3600 seconds
    Global Overrides Status: Enabled
    Global Filters Status: Enabled

IPS Auto Update is not currently configured

IPS Syslog and SDEE Notification Status
    Event notification through syslog is enabled
    Event notification through SDEE is enabled

IPS Signature Status
    Total Active Signatures: 1
    Total Inactive Signatures: 0

IPS Packet Scanning and Interface Status
    IPS Rule Configuration
      IPS name iosips
    IPS fail closed is disabled
    IPS deny-action ips-interface is false
    Fastpath ips is enabled
    Quick run mode is enabled
    Interface Configuration
      Interface GigabitEthernet0/1
        Inbound IPS rule is not set
        Outgoing IPS rule is iosips

IPS Category CLI Configuration:
    Category all
          Retire: True
    Category ios_ips basic
          Retire: False
R1#
```

rule is applied to interface GigabitEthernet0/1 in the Outbound direction.

3. Verify that IPS is working properly.
   a. From **PC-C**, attempt to ping **PC-A**. Were the pings successful? Explain.

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

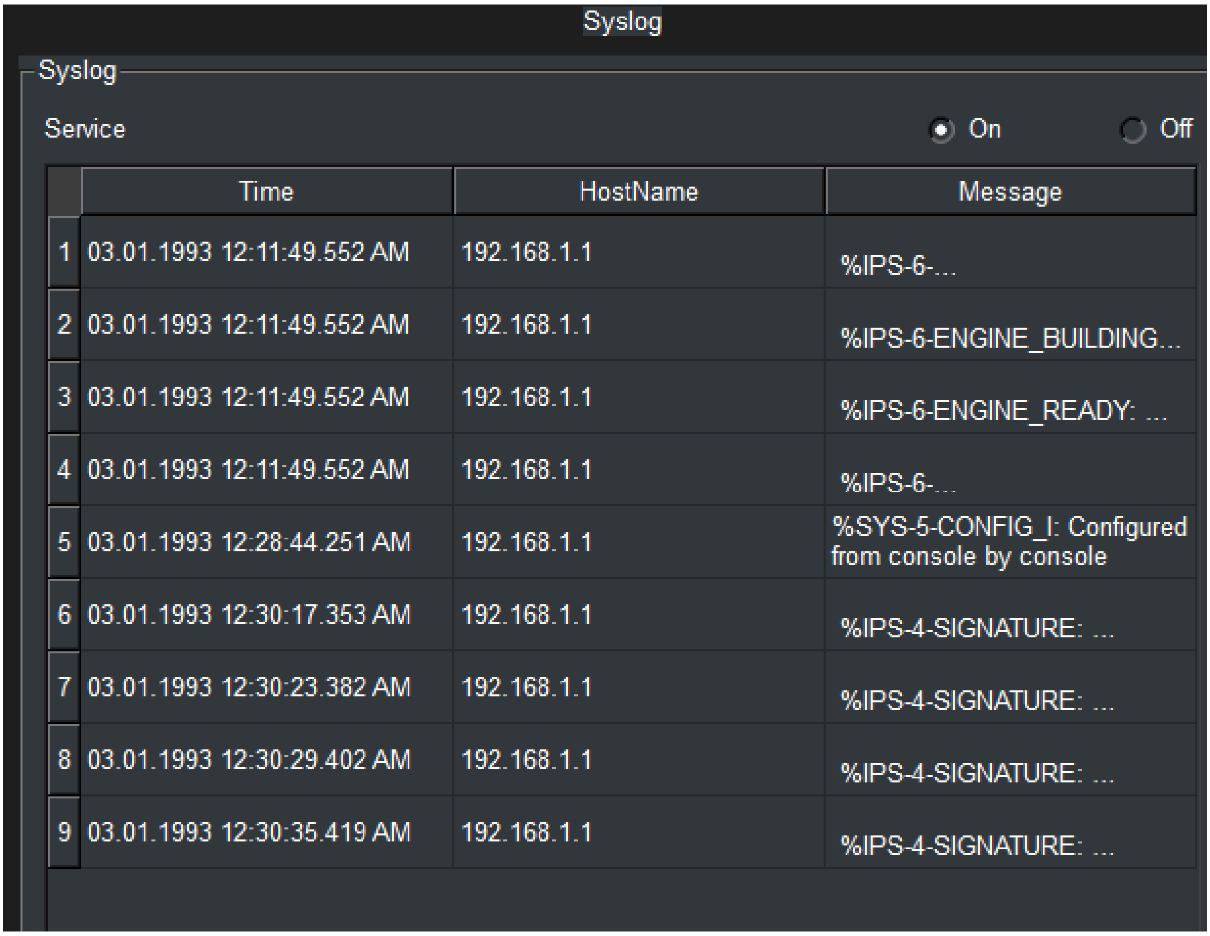
No, the pings fail. PC-C sends an ICMP Echo Request. As this request travels from R1 toward PC-A, it exits out of the G0/1 interface. Because our IPS rule inspects outbound traffic on G0/1 and we configured signature 2004 (echo request) to drop packets, the router intercepts and drops the request.

b. From **PC-A**, attempt to ping **PC-C**. Were the pings successful? Explain.

```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=2ms TTL=125
Reply from 192.168.3.2: bytes=32 time=3ms TTL=125
Reply from 192.168.3.2: bytes=32 time=3ms TTL=125
Reply from 192.168.3.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

Yes, the pings succeed. The initial Echo Request from PC-A enters G0/1 (inbound). Because our rule only checks outbound traffic, it passes. PC-C then responds with an Echo Reply. This reply exits G0/1 outbound and is inspected. However, because it is an Echo Reply and not an Echo Request (signature 2004), it does not trigger the drop action and successfully reaches PC-A.

4. View the syslog messages.
   a. Click the **Syslog** server.
   b. Select the **Services** tab.
   c. In the navigation menu, select **SYSLOG** to view the log file.
5. Check results.



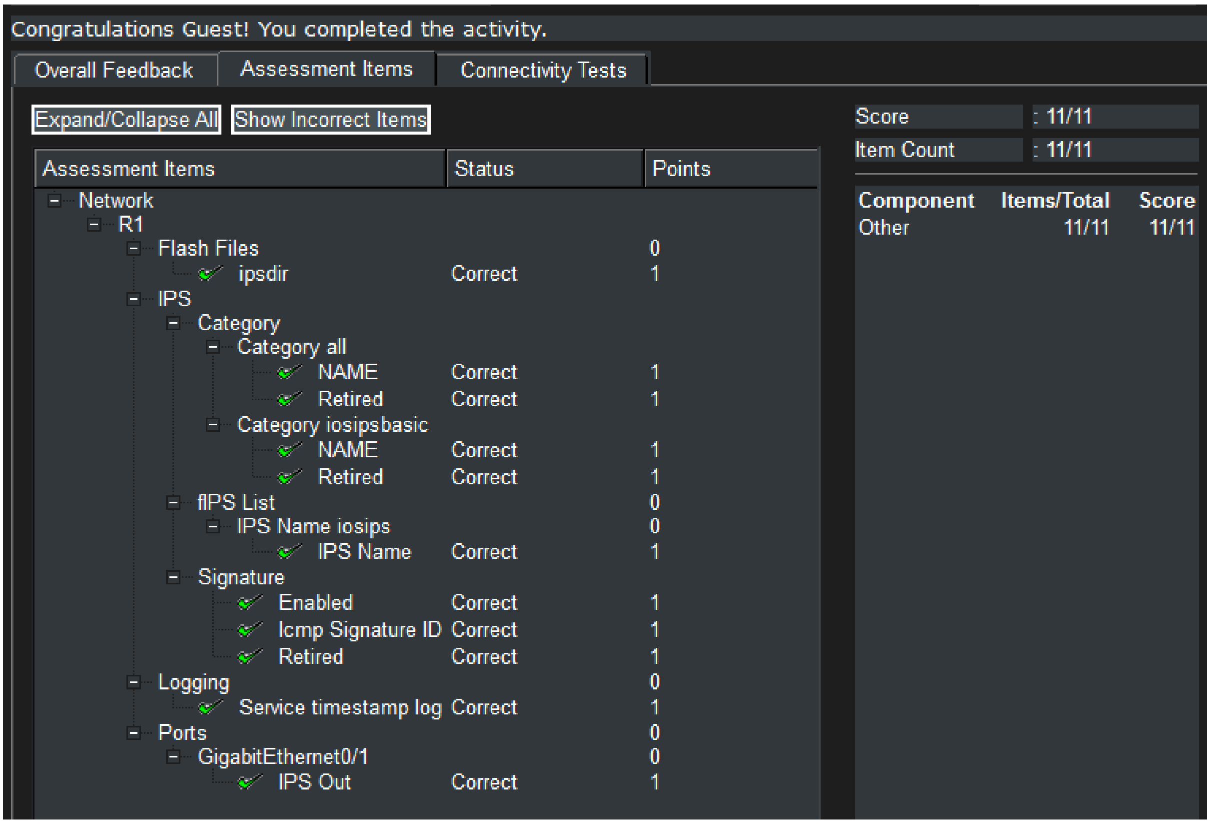
Syslog

Syslog

Service On Off

	Time	HostName	Message
1	03.01.1993 12:11:49.552 AM	192.168.1.1	%IPS-6-...
2	03.01.1993 12:11:49.552 AM	192.168.1.1	%IPS-6-ENGINE_BUILDING...
3	03.01.1993 12:11:49.552 AM	192.168.1.1	%IPS-6-ENGINE_READY: ...
4	03.01.1993 12:11:49.552 AM	192.168.1.1	%IPS-6-...
5	03.01.1993 12:28:44.251 AM	192.168.1.1	%SYS-5-CONFIG_I: Configured from console by console
6	03.01.1993 12:30:17.353 AM	192.168.1.1	%IPS-4-SIGNATURE: ...
7	03.01.1993 12:30:23.382 AM	192.168.1.1	%IPS-4-SIGNATURE: ...
8	03.01.1993 12:30:29.402 AM	192.168.1.1	%IPS-4-SIGNATURE: ...
9	03.01.1993 12:30:35.419 AM	192.168.1.1	%IPS-4-SIGNATURE: ...

Your completion percentage should be 100%. Click **Check Results** to see feedback and verification of which required components have been completed.



Task 2: Intrusion Detection with Snort

In this section, you will use an interactive lab environment to learn how to detect real-time threats and analyze network traffic using Snort.

1. Environment Setup

- **Access the Room:** Go to the TryHackMe Snort Room.
- **Account:** If you do not have an account, sign up using your academic or personal email.
- **Deploy VM:** Click the green **"Start Machine"** button in Task 2. Once the machine is ready, use the "Show Split View" to access the Linux terminal.

2. Lab Objectives Follow the instructions provided in the TryHackMe tasks to complete the following:

- **Task 2 & 4:** Familiarize yourself with the Snort CLI and verify the installation using snort -V.
- **Task 5 & 6:** Practice **Sniffer Mode** and **Packet Logger Mode** to view live headers and payloads.
- **Task 7 & 8:** Switch to **IDS/IPS Mode**. Use the provided traffic-generator.sh script to trigger alerts and investigate the log files.
- **Task 9 & 10:** Apply your knowledge to write custom Snort rules (e.g., filtering by IP ID, TCP flags, or specific port traffic) and run them against the provided .pcap files.

3. Submission Requirements

- **Completion:** Answer all the questions in each task on the TryHackMe platform to reach 100% progress.
- **Evidence:** * **Screenshot 1:** A capture of your terminal showing a successful Snort rule execution (e.g., the output of Task 9).

- **Screenshot 2:** Attach a screenshot of the completed TryHackMe room page showing the "Room Completed" banner or your 100% progress bar.

Assessment Rubric

Lab 08

Configuring IOS Intrusion Prevention System (IPS)

Name:	**Student ID:**

<u>Points Distribution</u>

Task No.	LR 2 Simulation	LR5 Results/Plots	LR9 Report

Task 1	30	15	
Task 2	30	15	
Total	/60	/30	/10
CLO Mapped	CLO 3	CLO 3	CLO3

Affective Domain Rubric		**Points**	**CLO Mapped**
AR 7	Report Submission	/10	CLO 3

CLO	**Total Points**	**Points Obtained**
3	100	
Total	**100**	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3:Good Points 3	Level 4:Exemplary Points 4
LR2	**Program/Code / Simulation Model/ Network Model**	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR9	**Report**	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.